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The Editors
Journal of Cheminformatics
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Dear Editors,

We are pleased to submit our manuscript, **"Larger models, unchanged verdict: a controlled benchmark of graph neural networks and transformers against ECFP4 fingerprints for antimalarial natural-product activity prediction"**, for consideration as a Research Article in the *Journal of Cheminformatics*.

The "scale-centric" narrative in molecular machine learning holds that larger pretrained models should supersede compact cheminformatics baselines. We test this assumption on a frozen panel of 19 836 African antimalarial natural products under both random and Bemis–Murcko scaffold cross-validation (25 fold–seed replicates), comparing a random forest on ECFP4 fingerprints against a graph isomorphism network (GIN), two topological-fusion variants (GIN–TFP, GIN–TNE), and a fine-tuned ChemBERTa transformer.

Under the scaffold split, the practically relevant out-of-distribution setting, **ECFP4–RF achieves the highest ROC AUC (0.830 0)**, ahead of GIN–TFP (0.813 8), GIN–TNE (0.809 0), GIN (0.804 7) and ChemBERTa (0.786 7); no graph or transformer arm significantly outperforms the fingerprint baseline. We document and correct a subtle cross-fold weight-leakage artifact in transformer fine-tuning that would otherwise inflate reported AUCs by up to 0.15, and an attribution analysis shows that persistent-image topological features carry the majority of the predictive signal, linking learned representations to molecular geometry even when predictive gains are absent.

We believe this controlled, leak-audited, fully reproducible benchmark aligns closely with the scope of the *Journal of Cheminformatics*—including its collection on the evaluation of AI and machine learning models in cheminformatics—and contributes an honest-negative result on a clinically motivated natural-product panel, mirroring the emerging consensus that representation–task alignment, not model scale, governs small-molecule activity prediction. Code, frozen splits and trained checkpoints are released under an open licence and are available to reviewers in the public repository (`https://github.com/NanaEngo/Malaria_codesV2`).

This work is original, is not under consideration elsewhere, and all authors have approved the submission.

We thank you for your consideration.

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Suggested reviewers:

- Prof. Gisbert Schneider (generative molecular design, fingerprints)
- Dr. Ola Engkvist (de novo design, graph models)
- Dr. Davide Boldini (natural-product fingerprints and chemical space)